

Uniformly bounded blueshift and continuation at the centre

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Abstract

We prove a continuation criterion for the spherically symmetric Einstein–scalar field equations. If the complete blueshift integrals are bounded uniformly on the incoming null cones in the past of a central endpoint, then $2m/r$ tends to zero uniformly throughout its past triangle. Christodoulou’s extension theorem then gives a C^1 continuation. The solution is assumed to be C^1 in Christodoulou’s sense before the endpoint, with C^1 characteristic datum α . The uniform bound need not be small. The result does not cover the assumption that the blueshift is finite only on the past light cone of the endpoint, nor does the proof establish the criterion for arbitrary absolutely continuous characteristic data.

Formal verification. The formal counterpart of the main theorem has been verified in Lean 4, with the cited Christodoulou BV inputs retained as explicit hypotheses. The manuscript and formalization were developed with OpenAI Codex under the prompts and direction of Serban Eugen Cicortas. The [versioned proof repository](#) and verification details are given in Appendix A. Repository access is currently restricted.

1 Introduction and statement

Christodoulou’s extension theorem gives continuation across a central endpoint if the mass ratio $2m/r$ is sufficiently small in a whole past triangle [1, p. 1172, Theorem 5.1 and (5.1)]. Finiteness of the blueshift on the single incoming cone ending at that point implies decay of the mass ratio along that cone [2, p. 190, Lemma 2 and (2.3)]. These are different smallness statements. The purpose of this note is to obtain the required full-triangle decay under a uniform bound for the blueshift on all incoming cones in the past region.

A related extension criterion is Singh’s [5, p. 200, Theorem 15], presented there as joint work with Xinliang An and Haoyang Chen. It assumes a uniform bound for the incoming blueshift integrals and the additional restriction $\sup \mu \leq 3/8 - \delta$. The scalar comparison below adapts Proposition 31 of that work. The continuation argument here is proved separately and does not impose that additional restriction.

1.1 Coordinates and regularity

We use $\text{Ric} = 2 \, d\phi \otimes d\phi$ and the notation of [1, pp. 1131, 1135–1137, (1.1c), (1.8), (1.11)–(1.17)]:

$$\begin{aligned} \lambda &= \partial_v r, & \nu &= \partial_u r, & \mu &= \frac{2m}{r}, \\ \theta &= r \partial_v \phi, & \zeta &= r \partial_u \phi, & \alpha &= \partial_v(r\phi) = \lambda\phi + \theta. \end{aligned} \tag{1}$$

Choose the double-null coordinates once, with matching labels on the regular axis, the endpoint at $(0, 0)$, and initial radius gauge

$$r(u_i, v) = \frac{v - u_i}{2}, \quad \lambda(u_i, v) = \frac{1}{2}.$$

These conditions are compatible: the axis labels can be reparametrized simultaneously using the initial radius, and then translated to put the endpoint at zero; compare [1, pp. 1135–1136]. The region is

$$\mathcal{Q} = \{(u, v) : u_i \leq u < 0, u \leq v \leq 0\}, \quad \Gamma = \{u = v\}.$$

The coordinates (u, v) and the meanings of $\nu, \lambda, \zeta, \theta$ remain fixed. Section 3 uses one explicitly named auxiliary chart (s, t) solely to prove a boundary estimate; its conclusion is expressed using a boundary measure independent of that chart. In particular, u is never rescaled along a selected ray or when a smaller cylinder is considered. We assume a solution with a regular centre on this region, with

$$\lambda > 0, \quad \nu < 0, \quad 0 \leq \mu < 1.$$

On every compact regular subdomain, including its axis boundary, $\lambda, \nu, \phi, \theta, \zeta$ are C^1 . This is the local regularity class in [1, p. 1157, definition of a C^1 solution]; its bounds $\inf \lambda > 0$ and $\inf \nu > -\infty$ hold on these compact subdomains. They are not assumed uniformly up to $(0, 0)$. The characteristic datum α on $u = u_i$ is C^1 on a neighborhood of $u_i \leq v \leq 0$ in this fixed initial gauge. Its endpoint on $v = 0$ has positive radius. No regularity at the missing central endpoint is assumed.

At fixed u , write

$$\partial_r = \lambda^{-1} \partial_v. \tag{2}$$

The radial derivative of α/λ used below is

$$\alpha' = \partial_r \left(\frac{\alpha}{\lambda} \right), \tag{3}$$

exactly as defined in [1, p. 1173, immediately before (5.2)]. In particular,

$$\frac{\alpha}{\lambda} = \phi + \frac{\theta}{\lambda}, \quad r \partial_r \left(\frac{\theta}{\lambda} \right) = r \alpha' - \frac{\theta}{\lambda}.$$

A partial u -derivative is taken at fixed v unless a fixed-radius derivative is expressly indicated. A derivative of a boundary value, such as $d\psi(u, L)/du$, is along $r = L$.

Theorem 1.1. *Under these assumptions, suppose that*

$$\sup_{u_i < v < 0} \int_{u_i}^v \frac{\mu(-\nu)}{(1-\mu)r} (u, v) du < \infty. \tag{4}$$

Then

$$\lim_{u \uparrow 0} \sup_{u \leq v \leq 0} \mu(u, v) = 0. \tag{5}$$

The solution consequently admits a C^1 continuation across $(0, 0)$, by [1, p. 1172, Theorem 5.1].

The integral in (4) runs from the initial outgoing cone to the regular axis along $v = \text{constant}$. It is unchanged by an increasing reparametrization of u . Its relation to the null expansion is the exact identity

$$\lambda(u, v) = \lambda(u_i, v) \exp \left(- \int_{u_i}^u \frac{\mu(-\nu)}{(1-\mu)r} (s, v) ds \right).$$

Thus the hypothesis gives a uniform bound for the corresponding exponential factors. The corresponding characterization of a positive lower bound for λ is stated in [1, p. 1147, following the definition of a BV solution]. The hypothesis also bounds the improper integral on $v = 0$; the proof of this assertion is included below. Uniform decay of the integral tails on the other cones is not an assumption.

1.2 Strategy and relation to the BV argument

The organization from geometric bounds through scalar nonconcentration to continuation parallels [3, Section 1.3.1 and Sections 3.3–3.5]. The estimates below are for the four-dimensional system; no circular-symmetry estimate is imported.

The proof has three parts. First, the uniform integral bound gives positive lower bounds for λ and $1 - \mu$. A comparison on a thin strip adjacent to an arbitrary incoming ray then bounds θ/λ . This adapts Singh’s running-maximum strip bootstrap [5, pp. 205–209, Proposition 31, especially (7.27)–(7.29)]: control the scalar field near a cone where it is large and use the mass equation to rule out excessive concentration. Singh attributes the underlying bootstrap to Liu–Li [6] in footnote 1 on p. 205. Related comparison-and-concentration arguments occur in [2, pp. 204–205, (2.63a)–(2.64) and (2.70b)–(2.72)]. Here the comparison is made inside an arbitrary incoming cone, without normalizing its outgoing coordinate, and the constants are uniform in that cone. The differentiated equation for α' is Christodoulou’s equation [1, p. 1173, (5.2)]. Its integrating factor, together with the uniform blueshift bound, gives a bound for $r\alpha'$. Integrating the radial mass equation on an interval where θ/λ is comparable to its value at the right endpoint then gives

$$\left| \frac{\theta}{\lambda} \right|^3 \leq C\mu.$$

This is the same interpolation principle used in [1, p. 1175, (5.7)–(5.8)]. The hypotheses used to close the estimates differ: in [1, p. 1172, (5.1)] smallness of μ on the whole triangle is already available. Here that smallness is the conclusion still to be proved.

Second, on a cylinder $0 \leq r \leq b$, we estimate its boundary energy in terms of the scalar data on $r = b$ during a time interval of length comparable to b . Elapsed time on that boundary is measured by $d\tau_b = (-\nu)(u, b)du$; this does not change u . The estimate is uniform even when $m(u, b)/b$ is arbitrarily small. It yields decay, on successive intervals, of

$$\frac{b}{2} \left[\left(\frac{\theta}{\lambda} \right)_b^2 - \log(1 - \mu_b) \right]$$

up to an error controlled by the variation of $(\alpha/\lambda)_b$. The uniform boundary estimate in Section 3 is proved here; it is an additional argument, not an invocation of a result from the BV paper.

Finally, finite blueshift on $v = 0$ gives $m(u, 0) = o(-u)$. If (5) failed, radial mass monotonicity would confine all points with $\mu > \varepsilon$ to radii $o(-u)$. A dyadic selection produces a cylinder whose duration divided by its radius is arbitrarily large and whose exterior has $\mu \leq \varepsilon$. The equation for $r\alpha'$ makes the variation of $(\alpha/\lambda)_b$ small after a specified waiting time. The boundary energy then decays below the mass required by a point with $\mu > \varepsilon$ inside that cylinder. This contradiction proves the full-triangle limit needed for Theorem 5.1.

Constants in the proof may depend on the uniform integral bound and on the fixed initial data. The constants in the local boundary estimates depend only on the two geometric constants specified there. They are independent of the cylinder radius and of the small mass threshold used in the final contradiction.

1.3 The boundary identity driving the argument

The principal identity is an exact balance law on a fixed-radius boundary. It is useful to state and derive it before the estimates. On $r = b$, define

$$Q_b(u) = \frac{b}{2} \left[\left(\frac{\theta}{\lambda} \right)_b^2 - \log(1 - \mu_b) \right]. \quad (6)$$

Here and throughout this subsection, every ordinary derivative is tangential to $r = b$ and uses the original coordinate u . The identity is

$$\boxed{\frac{dQ_b}{du} + \frac{b^2}{2(-\nu_b)} \left(\frac{d\phi_b}{du} \right)^2 = b \left(\frac{\theta}{\lambda} \right)_b \frac{d}{du} \left(\frac{\alpha}{\lambda} \right)_b.} \quad (7)$$

Its three terms are the change of boundary energy, a nonnegative scalar flux, and a term controlled by the variation of the trace $(\alpha/\lambda)_b$. No smallness assumption is used in this identity.

The cancellation. Since $v'_b = -\nu_b/\lambda_b$, the field equations give

$$\begin{aligned} b \frac{d\phi_b}{du} &= \zeta_b - \nu_b \left(\frac{\theta}{\lambda} \right)_b, \\ \frac{dm_b}{du} &= \frac{1 - \mu_b}{2} \left[\frac{\zeta_b^2}{\nu_b} - \nu_b \left(\frac{\theta}{\lambda} \right)_b^2 \right]. \end{aligned}$$

The logarithm in (6) is chosen precisely because

$$\frac{d}{du} \left[-\frac{b}{2} \log(1 - \mu_b) \right] = \frac{1}{1 - \mu_b} \frac{dm_b}{du}.$$

It cancels the factor $1 - \mu_b$ in the mass flux. Next use $(\theta/\lambda)_b = (\alpha/\lambda)_b - \phi_b$. The terms left after differentiating its square combine exactly as

$$\begin{aligned} \frac{1}{1 - \mu_b} \frac{dm_b}{du} - b \left(\frac{\theta}{\lambda} \right)_b \frac{d\phi_b}{du} &= \frac{1}{2\nu_b} \left[\zeta_b - \nu_b \left(\frac{\theta}{\lambda} \right)_b \right]^2 \\ &= -\frac{b^2}{2(-\nu_b)} \left(\frac{d\phi_b}{du} \right)^2. \end{aligned}$$

This proves (7), including its sign and lapse factor, directly in the fixed chart.

Why this is the right energy. It controls both quantities needed at the end of the proof:

$$Q_b \geq m_b, \quad Q_b \geq \frac{b}{2} \left(\frac{\theta}{\lambda} \right)_b^2,$$

because $-\log(1 - \mu) \geq \mu$. It is also unchanged by adding a constant to ϕ . The right side of (7) involves the derivative of α/λ , whose total variation becomes small on the selected cylinder; the value of this trace itself need not be small. If that trace were constant, Q_b would be nonincreasing, with its loss exactly equal to the integrated nonnegative term on the left.

Why the identity still needs an estimate. Monotonicity alone would allow a positive energy to remain while the flux is very small. Section 3 excludes this possibility uniformly in the mass ratio: over a boundary interval with $\int(-\nu_b)du = Hb$, for a fixed sufficiently large H depending only on the geometric bounds, it controls Q_b by the scalar flux and the variation of $(\alpha/\lambda)_b$. Combined with (7), this gives

$$Q_b(a_1) \leq qQ_b(a_0) + Cb \left[\text{TV}_{[a_0, a_1]}(\alpha/\lambda)_b \right]^2, \quad 0 < q < 1.$$

This is the decay estimate used in the final contradiction. The uniform boundary estimate supplies the quantitative loss; the exact identity converts that loss into decay of an energy controlling the enclosed mass. Both ingredients are necessary.

1.4 The wave-map argument and the provenance of the identity

Christodoulou–Tahvildar–Zadeh’s wave-map argument informed the search for an estimate excluding concentration. Its logarithmic multiplier controls kinetic energy near the axis; a separate target-space argument controls spatial energy [4, Section 2, Lemma 2 and Corollary 2, pp. 38–40; Lemma 3, pp. 40–43]. A coupled logarithmic analogue was investigated during this work, but its remaining bulk terms did not have an established sign. That current is not used in the proof below.

The exact boundary balance (7) does enter the proof. Its derivation above comes directly from the Einstein–scalar field equations. The related energy $\mathcal{E}_c = r[(\phi - c)^2 - \log(1 - \mu)]/2$, for a fixed scalar constant c , is used in Section 3 to prove the uniform boundary estimate. Thus the boundary identity and the wave-map logarithmic multiplier have distinct mathematical roles. We neither quote the boundary identity from that paper nor identify it with an unpublished identity.

1.5 Why C^1 data suffice, and the limitation for AC data

The differentiated transport equation used below does not require C^2 regularity of the datum α . On any compact characteristic rectangle in $r > 0$, $1 - \mu > 0$, the first-order fields and m are C^1 , and

$$\frac{\alpha}{\lambda}(u, v) = \frac{\alpha}{\lambda}(u_0, v) - \int_{u_0}^u \frac{\mu\nu}{(1 - \mu)r} \frac{\theta}{\lambda}(s, v) ds.$$

The integrand has a continuous v -derivative. Difference quotients therefore converge uniformly on a smaller compact rectangle, so this identity can be differentiated under the integral in v . The resulting identity is continuously differentiable in u . This supplies the mixed derivative used to derive the equation for α' , without assuming any pure second derivative of α . The remaining estimates use first derivatives, characteristic integration, and continuity on compact regular sets. For example, $\psi = r(\phi - c)$ has C^1 first derivatives, since $\psi_v = \alpha - c\lambda$ and $\psi_u = \nu(\phi - c) + \zeta$. The boundary energy identities and the equation for ψ_{uv} are consequently classical. At the regular axis, $\theta/\lambda = r\phi_r = O(r)$, $m = O(r^3)$, and $r\alpha' \rightarrow 0$ locally. None of these local assertions assumes uniform derivative bounds at the missing endpoint.

The initial derivative bound needed in Section 2 is

$$\sup_{u_i \leq v \leq 0} \left| \frac{r}{\lambda} \partial_v \left(\frac{\alpha}{\lambda} \right) (u_i, v) \right| < \infty.$$

The stated C^1 hypotheses give this bound. In Christodoulou’s initial gauge $u = 0$, $r = v/2$, writing the datum as $\alpha_0(v)$, the expression inside the absolute value is $2v d\alpha_0/dv$. Absolute continuity alone does not control it. For instance, a datum $\alpha_0(v) = \chi(v)|v - v_c|^\beta$, where $v_c > 0$, $0 < \beta < 1$, and χ is a smooth cutoff equal to one near v_c , is absolutely continuous with finite total variation but has an unbounded weighted derivative there. The amplitude can be arbitrarily small. Thus the present estimates cannot be passed to general AC data merely by approximation: the constants depending on this initial derivative need not remain bounded. This is a limitation of the proof, not a counterexample to an AC continuation criterion. The final theorem we invoke is itself a C^1 theorem [1, p. 1172, Theorem 5.1]. No optimality of C^1 regularity is claimed here.

2 Uniform geometric and scalar estimates

The field equations [1, p. 1137, (1.14)–(1.16)] take the form

$$r_u = \nu, \quad r_v = \lambda, \quad \lambda_u = \frac{\mu\lambda\nu}{(1-\mu)r}, \quad \nu_v = \frac{\mu\lambda\nu}{(1-\mu)r}, \quad (8)$$

$$m_u = \frac{(1-\mu)\zeta^2}{2\nu}, \quad m_v = \frac{(1-\mu)\theta^2}{2\lambda}, \quad m_r = \frac{1-\mu}{2} \left(\frac{\theta}{\lambda} \right)^2, \quad (9)$$

$$\theta_u = -\frac{\lambda\zeta}{r}, \quad \zeta_v = -\frac{\nu\theta}{r}, \quad \left(\frac{\alpha}{\lambda} \right)_u = -\frac{\mu\nu}{(1-\mu)r} \frac{\theta}{\lambda}. \quad (10)$$

The last identity is also [1, p. 1164, (4.10a)]. At the regular axis, $r = m = \theta = \zeta = 0$, and we set $\mu = 0$ by continuity. We shall also use the following consequences of these equations:

$$(\log(1-\mu))_u = -\frac{\zeta^2}{\nu r} + \frac{\mu\nu}{(1-\mu)r}, \quad (11)$$

$$\nu_r = \frac{\mu\nu}{(1-\mu)r}, \quad \left(\log \frac{-\nu}{1-\mu} \right)_r = \frac{1}{r} \left(\frac{\theta}{\lambda} \right)^2, \quad (12)$$

$$\partial_r \left(\frac{\mu\nu}{(1-\mu)r} \right) = \frac{\nu}{(1-\mu)r^2} \left[\left(\frac{\theta}{\lambda} \right)^2 - 2\mu \right]. \quad (13)$$

Expressions involving division by r are used for $r > 0$ and extended to the regular axis by their regular-axis limits whenever needed. Fix $B_* < \infty$ such that

$$\sup_{u_i < v < 0} \int_{u_i}^v \frac{-\mu\nu}{(1-\mu)r} (u, v) \, du \leq B_*. \quad (14)$$

Lemma 2.1 (Geometric estimates). *There are constants $\ell, \Lambda, \kappa > 0$ such that*

$$\ell \leq \lambda \leq \Lambda, \quad \kappa \leq 1 - \mu \leq 1. \quad (15)$$

In the gauge $u = v$ on the axis,

$$-\nu \geq \ell, \quad \ell(-u) \leq r(u, 0) \leq \Lambda(-u). \quad (16)$$

Set $K = \Lambda/\ell$. Let $u_i \leq a_0 < a_1 < 0$ and $0 < b \leq r(a_1, 0)$. Define $v_b(u)$ and $\tau(u)$ on $[a_0, a_1]$ by

$$r(u, v_b(u)) = b, \quad \tau(u) = \int_{a_0}^u [-\nu(s, v_b(s))] \, ds.$$

For $u_0 \in [a_0, a_1]$ put $v_0 = v_b(u_0)$. Then

$$\frac{\lambda(u_0, v_0)}{\lambda(u, v_0)} \leq K \quad \text{if } u_0 \leq u \leq \min\{a_1, v_0\}.$$

Moreover,

$$\tau(a_1) - \tau(u_0) \geq Kb \implies v_0 \leq a_1, \quad \tau(v_0) - \tau(u_0) \leq Kb.$$

In this conclusion, (v_0, v_0) is the regular axis endpoint of the incoming ray $v = v_0$. Thus the last assertion specifies both the available time interval and the time required to reach the axis.

Proof. The initial values of λ and $1 - \mu$ have positive lower bounds. Since $-\zeta^2/(\nu r) \geq 0$, integration of (8) and (11) gives

$$\lambda(u_i, v)e^{-B_*} \leq \lambda(u, v) \leq \lambda(u_i, v), \quad 1 - \mu(u, v) \geq [1 - \mu(u_i, v)]e^{-B_*}.$$

These bounds also hold on $v = 0$ by continuity on compact regular segments. More explicitly, fix $a \in (u_i, 0)$ and take $v_n \uparrow 0$ with $v_n > a$. Fatou's lemma and (14) give

$$\int_{u_i}^a \frac{-\mu\nu}{(1-\mu)r}(u, 0) du \leq \liminf_{n \rightarrow \infty} \int_{u_i}^a \frac{-\mu\nu}{(1-\mu)r}(u, v_n) du \leq B_*.$$

Letting $a \uparrow 0$ and using monotone convergence yields

$$\int_{u_i}^0 \frac{-\mu\nu}{(1-\mu)r}(u, 0) du \leq B_*.$$

Differentiating $r(u, u) = 0$ gives $-\nu = \lambda$ on the axis. The equation for ν_r implies that $-\nu$ is nondecreasing in r , and therefore $-\nu \geq \ell$. Moreover,

$$r(u, 0) = \int_u^0 \lambda(u, v) dv,$$

which proves (16).

The bound for $\lambda(u_0, v_0)/\lambda(u, v_0)$ follows from (15). The function v_b exists uniquely because $r(u, \cdot)$ increases from 0 to $r(u, 0) \geq b$. Also

$$v'_b(u) = \frac{-\nu(u, v_b(u))}{\lambda(u, v_b(u))}, \quad \tau'(u) = -\nu(u, v_b(u)), \quad d\tau = \lambda(u, v_b(u)) dv_b.$$

While $u < v_0$, the ray $v = v_0$ has not reached the axis, and

$$\tau(u) - \tau(u_0) \leq \Lambda[v_b(u) - v_0] < \Lambda[v_b(u) - u] \leq \frac{\Lambda}{\ell}b = Kb.$$

The last inequality follows from $b = \int_u^{v_b(u)} \lambda(u, v) dv \geq \ell[v_b(u) - u]$. If $\tau(a_1) - \tau(u_0) \geq Kb$, this strict inequality rules out $a_1 < v_0$. Letting $u \uparrow v_0$ then proves the asserted bound for $\tau(v_0) - \tau(u_0)$. $\square$

Lemma 2.2 (Scalar-field estimates). *There is a constant C such that*

$$\left| \frac{\theta}{\lambda} \right| + \left| \frac{\alpha}{\lambda} \right| + |\phi| \leq C. \quad (17)$$

Proof. The proof adapts the running-maximum bootstrap of [5, pp. 205–209, Proposition 31, especially (7.27)–(7.29)]; Singh credits Liu–Li in footnote 1 on p. 205. We apply the comparison to every incoming ray, retain the original null coordinates and their lapse weights, and make the constants uniform even for rays approaching the initial vertex. We keep the fixed null coordinates throughout. Choose any $v_* \in (u_i, 0]$, and let a subscript 0 denote restriction to the incoming ray $v = v_*$. Thus

$$r_0(u) = r(u, v_*), \quad \frac{dr_0}{du} = \nu_0(u) < 0.$$

We use this decreasing radius as an integration variable on the selected ray; we do not change the outgoing coordinate. The initial transverse Lipschitz bounds for λ and θ are uniform on the compact initial segment by the C^1 hypotheses.

If $d = v_* - v \geq 0$ and $\Lambda d \leq r_0(u)/2$, then

$$\frac{1}{2}r_0 \leq r \leq r_0, \quad 2^{-(1-\kappa)/\kappa} \leq \frac{\nu}{\nu_0} \leq 1. \quad (18)$$

Indeed, an axis point between v and v_* would imply $r_0 \leq \Lambda d$, which is impossible. Therefore the entire radial segment lies in the domain, and $0 \leq r_0 - r \leq \Lambda d$. Also

$$\log \frac{-\nu_0(u)}{-\nu(u, v)} = \int_{r(u, v)}^{r_0(u)} \frac{\mu}{(1-\mu)s} ds \in \left[0, \frac{1-\kappa}{\kappa} \log 2\right].$$

This proves the lapse-ratio bound. A strip satisfying the radius restriction at its final time satisfies it at all preceding times, since r_0 is decreasing.

On the selected ray, $\zeta_0 = r_0\phi_{u,0}$, so (11) gives the weighted energy identity

$$\int_{u_i}^u \frac{r_0|\phi_{u,0}|^2}{-\nu_0} dt = \log \frac{1-\mu_0(u)}{1-\mu_0(u_i)} + \int_{u_i}^u \frac{\mu_0(-\nu_0)}{(1-\mu_0)r_0} dt \leq B_* + \log(1/\kappa). \quad (19)$$

For $v_* = 0$, the last bound uses the limiting-cone integral already established in Lemma 2.1.

Fix $u_i \leq u_* < v_*$ and set

$$F = 1 + \max_{u_i \leq u \leq u_*} |\theta_0(u)|.$$

Choose $K_0 \geq 4$ larger than the uniform initial scalar bound. On

$$[u_i, u_*] \times [v_* - \eta, v_*], \quad \eta = \delta \frac{r_0(u_*)}{F},$$

we bootstrap $|\theta| \leq K_0 F$. Initially take $0 < \delta \leq (2\Lambda)^{-1}$. Since $F \geq 1$, this ensures the wedge condition on the whole rectangle. It also keeps its initial edge within the given data segment, uniformly in v_* : indeed,

$$\eta \leq \delta r_0(u_i) = \delta \int_{u_i}^{v_*} \lambda(u_i, v) dv \leq \delta \Lambda (v_* - u_i) \leq \frac{1}{2}(v_* - u_i).$$

Thus the rectangle is a compact regular subset of $\mathcal{Q}$.

Under the bootstrap assumption, the radial equations give

$$\begin{aligned} |r - r_0| &\leq Cd, & \left| \frac{\nu}{\nu_0} - 1 \right| &\leq C \frac{d}{r_0}, \\ |m - m_0| &\leq CK_0^2 F^2 d, & \frac{|\zeta - \zeta_0|}{-\nu_0} &\leq CK_0 F \frac{d}{r_0}. \end{aligned} \quad (20)$$

For the lapse comparison, divide $\nu_v = \mu\lambda\nu/[(1-\mu)r]$ by the fixed value $\nu_0(u)$ on each outgoing slice and integrate in v ; the ratio in (18) is at most one. For the last comparison use $|\zeta_v| = |\nu\theta|/r \leq 2(-\nu_0)K_0 F/r_0$. No time derivative of ν/ν_0 or ζ/ν_0 is used.

It follows, using $m_0 \leq r_0/2$ and $1-\mu \geq \kappa$, that

$$\begin{aligned} |\mu - \mu_0| + \left| \frac{\nu/\nu_0}{1-\mu} - \frac{1}{1-\mu_0} \right| &\leq CK_0^2 F^2 \frac{d}{r_0}, \\ \left| \frac{\mu\nu}{(1-\mu)r} - \frac{\mu_0\nu_0}{(1-\mu_0)r_0} \right| &\leq C(-\nu_0)K_0^2 F^2 \frac{d}{r_0^2}. \end{aligned}$$

The constants here are independent of K_0, F, u_*, v_* and of the size of $-\nu_0$. Integrating the difference of the equations for $\log \lambda$ uses the exact identity

$$\int_{u_i}^u \frac{-\nu_0(t)}{r_0(t)^2} dt = \frac{1}{r_0(u)} - \frac{1}{r_0(u_i)} \leq \frac{1}{r_0(u)}.$$

The initial Lipschitz difference is bounded by Cd and can be absorbed using the uniform upper bound for $r_0(u)$. Since $\ell \leq \lambda \leq \Lambda$, the mean-value theorem for the exponential function then gives

$$|\lambda(u, v) - \lambda_0(u)| \leq CK_0^2 F^2 \frac{d}{r_0(u)}. \quad (21)$$

This deduction does not require the right-hand side to be small.

Since $\phi_u = \zeta/r$ and $\theta_u = -\lambda\phi_u$, the preceding estimates imply

$$\begin{aligned} |\phi_u - \phi_{u,0}| &\leq CK_0 F \frac{(-\nu_0)d}{r_0^2} + C \frac{d}{r_0} |\phi_{u,0}|, \\ |\theta_u - \theta_{u,0}| &\leq CK_0^2 F^2 d \left(\frac{-\nu_0}{r_0^2} + \frac{|\phi_{u,0}|}{r_0} \right). \end{aligned}$$

The weight in (19) supplies exactly the factor needed to integrate the second term. Cauchy–Schwarz gives

$$\begin{aligned} \int_{u_i}^u \frac{|\phi_{u,0}|}{r_0} dt &\leq \left(\int_{u_i}^u \frac{r_0 |\phi_{u,0}|^2}{-\nu_0} dt \right)^{1/2} \left(\int_{u_i}^u \frac{-\nu_0}{r_0^3} dt \right)^{1/2} \\ &\leq \frac{C}{r_0(u)}, \end{aligned} \quad (22)$$

because

$$\int_{u_i}^u \frac{-\nu_0(t)}{r_0(t)^3} dt = \frac{1}{2} \left(\frac{1}{r_0(u)^2} - \frac{1}{r_0(u_i)^2} \right).$$

Integration of the θ_u comparison, including its uniform initial Lipschitz difference, therefore yields

$$|\theta(u, v) - \theta_0(u)| \leq CK_0^2 F^2 \frac{d}{r_0(u)}. \quad (23)$$

Fix K_0 , and reduce δ until $CK_0^2 \delta \leq 1/4$ for the constant in the last estimate. Throughout the rectangle,

$$CK_0^2 F^2 \frac{d}{r_0(u)} \leq CK_0^2 \delta F \frac{r_0(u_*)}{r_0(u)} \leq F/4.$$

To close the bootstrap, vary the width from zero to η while keeping the time interval fixed. The maximum of $|\theta|$ on the resulting rectangles is continuous in their width: parametrize each rectangle by $(u, v) = (u, v_* - w\sigma)$ with $u \in [u_i, u_*]$, $\sigma \in [0, 1]$, and use uniform continuity on this compact parameter set. The bootstrap holds strictly at width zero. At any width where it holds, (23) improves it to

$$|\theta| \leq F - 1 + F/4 < K_0 F.$$

The admissible widths consequently form a nonempty relatively open and closed subset of $[0, \eta]$, and therefore fill that interval. For a related continuity closure of a strip bootstrap, see [1, pp. 1177–1179, (5.13)–(5.21)].

Suppose now that u_* attains the maximum defining F and that $F \geq 2$. Then $|\theta(u_*, v)| \geq F/4$ throughout the final strip. The radial mass equation and the nonnegativity of the mass at the strip's left endpoint imply

$$m_0(u_*) \geq \int_{v_*-\eta}^{v_*} \frac{(1-\mu)\theta^2}{2\lambda} dv \geq \frac{\kappa\delta}{32\Lambda} F r_0(u_*).$$

Since $m_0(u_*) < r_0(u_*)/2$, this bounds F uniformly. For an arbitrary evaluation time, take a maximizer on its preceding compact time interval and apply the same argument with that maximizer as the final time; the value of F is unchanged. If the maximizer is the initial time, the fixed initial bound suffices. The case $F < 2$ is already bounded. Every $v_* \in (u_i, 0]$ was allowed, and regular-axis values follow by continuity. Thus θ , and hence θ/λ , are bounded uniformly on $\mathcal{Q}$.

Finally, integrate the last equation in (10) along each incoming ray. Its coefficient has absolute integral at most B_* , and θ/λ is now bounded. The initial values of α/λ are bounded, so α/λ is uniformly bounded as well. The identity $\phi = \alpha/\lambda - \theta/\lambda$ completes the proof of (17). $\square$

Lemma 2.3 (A radial derivative estimate). *There is a constant C such that*

$$|r\alpha'| \leq C, \quad \left| \frac{\theta}{\lambda} \right|^3 \leq C\mu. \quad (24)$$

Proof. The commutator identity

$$[\partial_u, \partial_r] = -\frac{\mu\nu}{(1-\mu)r} \partial_r$$

together with (10) and (13) gives [1, p. 1173, (5.2)]

$$(r\alpha')_u = \frac{\nu}{r} \left(1 - \frac{2\mu}{1-\mu} \right) r\alpha' - \frac{\nu}{(1-\mu)r} \frac{\theta}{\lambda} \left[\left(\frac{\theta}{\lambda} \right)^2 - 3\mu \right]. \quad (25)$$

Along a fixed ingoing characteristic, the homogeneous propagator from t to u is

$$\frac{r(u, v)}{r(t, v)} \exp \left(2 \int_t^u \frac{-\mu\nu}{(1-\mu)r} (s, v) ds \right) \leq e^{2B_*} \frac{r(u, v)}{r(t, v)}.$$

The initial value of $r\alpha'$ is uniformly bounded, including at the initial axis point. Lemmas 2.1 and 2.2 control the remaining factors in the inhomogeneous term. Variation of constants therefore gives

$$\begin{aligned} |r\alpha'(u, v)| &\leq e^{2B_*} \frac{r(u, v)}{r(u_i, v)} |r\alpha'(u_i, v)| + C e^{2B_*} r(u, v) \int_{u_i}^u \frac{-\nu(t, v)}{r(t, v)^2} dt \\ &\leq C, \end{aligned} \quad (26)$$

where we used the exact identity

$$r(u, v) \int_{u_i}^u \frac{-\nu(t, v)}{r(t, v)^2} dt = 1 - \frac{r(u, v)}{r(u_i, v)} \leq 1. \quad (27)$$

Choose $C_0 \geq 1$ such that

$$\left| \frac{\theta}{\lambda} \right| + \left| r \partial_r \left(\frac{\theta}{\lambda} \right) \right| \leq C_0.$$

On a fixed outgoing cone, let $A = |\theta/\lambda|(r)$ at a point with $r > 0$, and set $d = rA/(8C_0)$. If $A = 0$, the second conclusion is immediate. Otherwise $d \leq r/8$, and for $t \in [r-d, r]$,

$$\left| \left(\frac{\theta}{\lambda} \right)(t) - \left(\frac{\theta}{\lambda} \right)(r) \right| \leq C_0 \log \frac{r}{t} \leq C_0 \frac{d}{r-d} \leq A/2.$$

Integrating the radial mass equation over this interval gives

$$m(r) \geq \int_{r-d}^r \frac{1-\mu}{2} \left(\frac{\theta}{\lambda}\right)^2 dt \geq \frac{\kappa}{64C_0} r A^3.$$

Since $\mu = 2m/r$, this proves the cubic estimate. This local-interval proof uses the same mass/derivative interpolation principle as [1, p. 1175, (5.7)–(5.8)]; it does not use that paper’s weighted estimate or its small-mass hypothesis. $\square$

3 A uniform boundary energy estimate

We shall prove that the mass inside $r = L$ is controlled by the scalar field on $r = L$ over a time interval proportional to L . The estimate will be uniform as the initial mass divided by L tends to zero. We first prove it with constants depending on a positive lower bound for this ratio. We then show that, if the boundary integral is small, most of the mass must remain inside a radius proportional to the initial mass. Applying the first estimate at that smaller radius removes the dependence on the initial mass ratio.

For a boundary $r = b$, let $v = v_b(u)$. Define its measure and its tangential derivative by

$$d\tau_b = (-\nu_b) du, \quad \frac{df_b}{d\tau_b} = \frac{f_u - \nu_b f_r}{-\nu_b} \Big|_{r=b}. \quad (28)$$

Here f_u is still the null derivative at fixed v , whereas f_b denotes the restriction to $r = b$. The formula follows from $v'_b = -\nu_b/\lambda_b$. Both the measure and derivative are unchanged by separate increasing reparametrizations of the null labels. For $\psi = r(\phi - c)$, the boundary integral is consequently

$$\int \left[\psi_r^2 + \left(\frac{d\psi_b}{d\tau_b} \right)^2 \right] d\tau_b = \int \left[(-\nu_b) \psi_r^2 + \frac{(\psi_u - \nu_b \psi_r)^2}{-\nu_b} \right]_{r=b} du. \quad (29)$$

A cylinder’s duration below means the integral of this measure on its outer boundary. All boundary estimates are statements about (29), with these durations.

For their proof only, fix a cylinder $0 \leq r \leq L$ between $u = a_0$ and $u = a_1$, and introduce auxiliary characteristic parameters

$$s(u) = \int_{a_0}^u [-\nu(w, v_L(w))] dw, \quad t(v) = s(v_L^{-1}(v))$$

on incoming rays entering through $r = L$. Extend t as a C^2 function with $t' > 0$ to earlier labels if needed. Bounds using an entry value below are asserted only on rays that have entered after a_0 . Put $T = s(a_1)$. The outer boundary is $t = s$. On entered rays,

$$\frac{ds}{du} = -\nu_L, \quad \frac{dt}{dv} = \lambda(v_L^{-1}(v), v), \quad r_s = \frac{\nu}{-\nu_L}, \quad r_t = \frac{\lambda(u, v)}{\lambda(v_L^{-1}(v), v)}.$$

In particular, $r_s = -1$ and $r_t = 1$ on $r = L$. We keep this one auxiliary chart throughout the boundary argument, including the estimates at smaller radii. We write $r_s, r_t, r\phi_s$ explicitly; they are not assigned the symbols ν, λ, ζ . The ratios $\theta/\lambda = r\phi_r$ and $\alpha/\lambda = (r\phi)_r$ retain their original meanings. On $r = L$, $d\tau_L = ds$, so the integrals and durations written in s below are exactly those just defined. For any smaller radius,

$$d\tau_b = (-r_s)_b ds = (-\nu_b) du.$$

Thus statements proved below apply at $r = b$ with that measure, without choosing another chart. A partial s -derivative is at fixed t ; a boundary value has derivative

$$\frac{d}{ds}\psi(s, b) = \psi_s(s, b) - r_s(s, b)\psi_r(s, b), \quad \partial_r = r_t^{-1}\partial_t.$$

We assume the conclusions of Lemma 2.1, with $K \geq 1$, in the following precise form:

$$1 - \mu \geq \kappa > 0, \quad \frac{r_t(a, t_b(a))}{r_t(s, t_b(a))} \leq K. \quad (30)$$

Here $r(s, t_b(s)) = b$, $0 < b \leq L$, and the second estimate holds along each ray $t = t_b(a)$ between its crossing of $r = b$ at time a and its meeting with the axis. Moreover, whenever

$$\int_a^T (-r_s)(s, b) ds \geq Kb,$$

there exists $\sigma \in [a, T]$ such that

$$r(\sigma, t_b(a)) = 0, \quad \int_a^\sigma (-r_s)(s, b) ds \leq Kb.$$

These assumptions apply to every smaller regular cylinder contained in the given one. Thus the same K bounds both the ratio in (30) and the time to the axis measured by $d\tau_b = (-r_s)_b ds$. All constants in this section depend only on κ, K , unless a mass-ratio threshold is indicated. Indeed, along a fixed incoming ray the factor dt/dv cancels:

$$\frac{r_t(s_0, t)}{r_t(s, t)} = \frac{\lambda(u_0, v)}{\lambda(u, v)}.$$

Together with $d\tau_b = (-\nu_b)du$, this proves that the hypotheses of Lemma 2.1 pass to this chart with exactly the same constants. No bound for ds/du is used.

Fix a constant c and set

$$\psi = r(\phi - c).$$

Then

$$\psi_r = \frac{\alpha}{\lambda} - c, \quad \frac{\theta}{\lambda} = \psi_r - \frac{\psi}{r}.$$

The equations (8)–(10) give

$$\begin{aligned} m_r &= \frac{1 - \mu}{2} \left(\frac{\theta}{\lambda} \right)^2, & m_s &= \frac{1 - \mu}{2r_s} (r\phi_s)^2, \\ \partial_r(r\phi_s) &= -\frac{r_s}{r} \frac{\theta}{\lambda}, & (r_s)_r &= \frac{\mu r_s}{(1 - \mu)r}, \\ r_{st} &= \frac{\mu r_s r_t}{(1 - \mu)r}, & \psi_{st} &= -V\psi, \quad V = -\frac{\mu r_s r_t}{(1 - \mu)r^2} \geq 0. \end{aligned} \quad (31)$$

In particular, along $r = L$,

$$\frac{d}{ds}\psi(s, L) = (r\phi_s)_L + \frac{\theta_L}{\lambda_L}, \quad \frac{d}{ds}m(s, L) = \frac{1 - \mu_L}{2} \left[\left(\frac{\theta_L}{\lambda_L} \right)^2 - (r\phi_s)_L^2 \right].$$

Define the boundary energy by

$$\mathcal{E}_c(s, L) = \frac{\psi(s, L)^2}{2L} - \frac{L}{2} \log(1 - \mu_L) \geq m(s, L). \quad (33)$$

The preceding identities imply the exact relation

$$\frac{d}{ds}\mathcal{E}_c(s, L) = \psi_r(s, L)\frac{d}{ds}\psi(s, L) - \frac{1}{2}\left(\frac{d}{ds}\psi(s, L)\right)^2. \quad (34)$$

Indeed, the derivative of the logarithmic term is $\frac{1}{2}[(\theta_L/\lambda_L)^2 - (r\phi_s)_L^2]$; substituting the first boundary identity gives (34). Whenever an interval I is specified, we write

$$E = \int_I \left[\psi_r(s, L)^2 + \left(\frac{d}{ds}\psi(s, L) \right)^2 \right] ds$$

for its boundary integral. There is only one such integral in each argument below, and its interval will be stated explicitly.

3.1 A positive lower bound for the initial mass ratio

The next argument uses lapse comparison and weighted Cauchy–Schwarz to control an annular mass difference, as in [3, Section 4, p. 35, (4.1)–(4.4)]. Here the outer boundary has fixed radius; the four-dimensional coefficients and the resulting Gronwall estimate are derived below.

Lemma 3.1. *Let $M_0 = m(0, L) > 0$ and set*

$$J_{\text{in}} = \int_0^T \frac{1 - \mu_L}{2} \left(\frac{\theta_L}{\lambda_L} \right)^2 ds, \quad J_{\text{out}} = \int_0^T \frac{1 - \mu_L}{2} (r\phi_s)_L^2 ds, \quad M_+ = M_0 + J_{\text{in}}.$$

Choose $2M_+ < b < L$. The incoming ray through $(0, L)$, written as $r = z(s)$, reaches b within time

$$(L - b) + 2M_+ \log \frac{L - 2M_+}{b - 2M_+},$$

provided the cylinder exists for that duration. Put

$$d(x) = 1 - \frac{2M_+}{x}, \quad H = 2 \int_b^L \frac{x^{-1} - L^{-1}}{d(x)^2} dx.$$

Until that ray reaches b ,

$$\begin{aligned} m(s, L) - m(s, z(s)) &\leq D_* := e^H \left[J_{\text{in}} + \left(\frac{2}{d(b)} - 1 \right) J_{\text{out}} \right], \\ m(s, z(s)) &\geq M_0 - D_* - J_{\text{out}}. \end{aligned} \quad (35)$$

Proof. Boundary mass balance and $m_r \geq 0$ give $0 \leq m \leq M_+$. Radial integration of the equation for r_s gives $d(r) \leq -r_s \leq 1$ for $r \geq b$. Also

$$\partial_r \log \frac{-r_s}{1 - \mu} = \frac{1}{r} \left(\frac{\theta}{\lambda} \right)^2 = \frac{2m_r}{r - 2m}.$$

Since $z > 2M_+$ and $m_r \geq 0$,

$$\frac{1 - \mu_z}{-(r_s)_z} = (1 - \mu_L) \exp \left(\int_z^L \frac{2m_r}{r - 2m} dr \right) \leq (1 - \mu_L) \frac{z - 2m_z}{z - 2m_L} \leq \frac{1 - \mu_L}{d(z)}.$$

The first inequality replaces $r - 2m$ by the smaller positive quantity $z - 2m$. The same radial identity gives

$$\frac{(-r_s)^2}{1 - \mu} \leq \frac{1}{1 - \mu_L}.$$

Since $z' = (r_s)_z$, integrating $-z' \geq d(z)$ proves the stated time bound.

Write $D(s) = m(s, L) - m(s, z(s))$. By Cauchy–Schwarz,

$$|(r\phi_s)_L - (r\phi_s)_z|^2 \leq 2D \int_z^L \frac{(-r_s)^2}{(1 - \mu)r^2} dr \leq \frac{2D}{1 - \mu_L} \left(\frac{1}{z} - \frac{1}{L} \right).$$

Together with $(r\phi_s)_z^2 \leq 2(r\phi_s)_L^2 + 2((r\phi_s)_z - (r\phi_s)_L)^2$, this implies

$$\frac{(1 - \mu_z)(r\phi_s)_z^2}{2(-(r_s)_z)} \leq \frac{1 - \mu_L}{d(z)} (r\phi_s)_L^2 + \frac{2(z^{-1} - L^{-1})}{d(z)} D.$$

Along the ray, $dm(s, z(s))/ds = m_s(s, t)$ with its incoming label fixed. Hence

$$D' \leq \frac{1 - \mu_L}{2} \left[\left(\frac{\theta_L}{\lambda_L} \right)^2 + \left(\frac{2}{d(z)} - 1 \right) (r\phi_s)_L^2 \right] + \frac{2(z^{-1} - L^{-1})}{d(z)} D.$$

The time integral of the last coefficient is at most H , since $-z' \geq d(z)$. Gronwall's inequality, $D(0) = 0$, and $d(z) \geq d(b)$ prove the first bound in (35). The second follows from $m(s, L) \geq M_0 - J_{\text{out}}$. $\square$

Lemma 3.2. *Suppose $\epsilon \leq 2m(0, L)/L \leq 1 - \kappa$. There are positive $C_\epsilon, e(\epsilon)$ such that every regular cylinder of duration $C_\epsilon L$ satisfies, for every constant c ,*

$$\int_0^{C_\epsilon L} \left[\psi_r(s, L)^2 + \left(\frac{d}{ds} \psi(s, L) \right)^2 \right] ds > e(\epsilon) L. \quad (36)$$

For sufficiently small $\epsilon > 0$ they may be chosen with

$$C_\epsilon \leq C(1 + |\log \epsilon|), \quad e(\epsilon) \geq \frac{\epsilon^3}{C(1 + |\log \epsilon|)^2}. \quad (37)$$

Proof. Let $M_0 = m(0, L)$ and $J = J_{\text{in}} + J_{\text{out}}$, with the integrals taken over the full cylinder. Set $\delta_0 = \kappa/8$ and $b = 2M_0(1 + \delta_0)$. Suppose $J \leq c_0 \epsilon^2 M_0$, where $c_0 > 0$ will be fixed using only κ . Taking $c_0 \leq \delta_0/2$ gives

$$b < L, \quad b - 2M_+ \geq \delta_0 M_0, \quad d(b) \geq \frac{\delta_0}{2(1 + \delta_0)}.$$

The first inequality follows from $b/L \leq (1 - \kappa)(1 + \kappa/8) < 1$. The integral in the preceding lemma obeys

$$H \leq 2 \log \frac{L}{\delta_0 M_0} + \frac{4}{\delta_0} + 2 \leq 2 \log \frac{2}{\epsilon \delta_0} + \frac{4}{\delta_0} + 2.$$

To see the coefficient of the logarithm explicitly, put $a = 2M_+$ and use

$$H \leq 2 \int_b^L \left[\frac{1}{x - a} + \frac{a}{(x - a)^2} \right] dx \leq 2 \log \frac{L}{\delta_0 M_0} + \frac{4(1 + c_0)}{\delta_0}.$$

Thus $e^H \leq C\epsilon^{-2}$, with C depending only on κ . It follows that $D_* + J_{\text{out}} \leq C\epsilon^{-2}J$. Fix c_0 still smaller so that this is at most $\delta_0 M_0$. The ray reaches b by time $[1 + \log(2/(\epsilon\delta_0))]L$, and at that point

$$1 - \mu(s, b) \leq 1 - \frac{1 - \delta_0}{1 + \delta_0} = \frac{2\delta_0}{1 + \delta_0} < \kappa.$$

Consequently

$$J > c_0 \epsilon^2 M_0 \geq \frac{c_0 \epsilon^3 L}{2}. \quad (38)$$

Choose

$$C_\epsilon = \max \left\{ 1 + \log \frac{2}{\epsilon\delta_0}, K + 1 \right\}, \quad \eta^2 = \frac{c_0 \epsilon^3}{96 C_\epsilon},$$

$$e(\epsilon) = \min \left\{ \frac{\eta^2}{16K}, \frac{c_0 \epsilon^3}{16(1 + 6C_\epsilon^2)}, K^{-1} \right\}.$$

Write E for the integral in (36) and suppose $E \leq e(\epsilon)L$. If $|\psi(0, L)| < \eta L$, Cauchy-Schwarz gives

$$\int_0^{C_\epsilon L} \left(\frac{\theta_L}{\lambda_L} \right)^2 ds \leq 2 \int \psi_r(s, L)^2 ds + 4C_\epsilon L \eta^2 + 4C_\epsilon^2 \int \left(\frac{d}{ds} \psi(s, L) \right)^2 ds.$$

Use $(r\phi_s)_L = d\psi(s, L)/ds - \theta_L/\lambda_L$ and $1 - \mu_L \leq 1$ to obtain

$$J \leq 3 \int \psi_r(s, L)^2 ds + (1 + 6C_\epsilon^2) \int \left(\frac{d}{ds} \psi(s, L) \right)^2 ds + 6C_\epsilon L \eta^2$$

$$\leq (1 + 6C_\epsilon^2)E + 6C_\epsilon L \eta^2 \leq \frac{c_0 \epsilon^3 L}{8},$$

Here $C_\epsilon \geq K + 1 \geq 2$ justifies the second inequality. The result contradicts (38).

Otherwise, reverse the sign of ψ if necessary and put $q = \psi(0, L) \geq \eta L$. Integrating $\psi_{st} = -V\psi$ first along an incoming ray and then along an outgoing slice gives

$$\psi(s, t) = \psi(s, L) - \int_t^s \psi_r(\sigma, L) d\sigma + \int_t^s \int_t^\sigma V(\sigma, z) \psi(\sigma, z) dz d\sigma. \quad (39)$$

Let $\rho = \kappa\eta^2/(64 + \kappa\eta^2)$, and follow the ray $t = 0$ until its radius is ρL . This takes at most KL . The first two terms in (39) are at least

$$q - \sqrt{2KLE} \geq q/2.$$

Every integration point satisfies $t \leq z \leq \sigma \leq s$ and $r(\sigma, z) \geq r(\sigma, t) \geq r(s, t) \geq \rho L$. The operator defined by the double integral therefore preserves this region and has a nonnegative kernel. On this compact region its n th iterate has norm at most $(T^2 \sup V)^n/n!$, by the ordered time integrations. Its Neumann series converges and preserves nonnegative functions. It follows that $\psi \geq q/2$ throughout the region.

The once-integrated equation now gives

$$(\psi_r)_+(s, t) \leq \frac{|\psi_r(t, L)|}{r_t(s, t)}, \quad \int_{\rho L}^L (\psi_r)_+^2 dr \leq KE$$

on the final outgoing slice; here $dr = r_t dt$ and $r_t^{-1} \leq K$. For real x and $a > 0$, $(x - a)^2 \geq a^2/2 - (x_+)^2$. Hence the mass equation yields

$$m(s, L) \geq \frac{\kappa q^2}{16L} (\rho^{-1} - 1) - \frac{K}{2} E \geq \frac{4q^2}{\eta^2 L} - \frac{L}{2} \geq \frac{7}{2} L,$$

contradicting $2m < L$. This proves (36). The displayed choices give (37), since c_0 is fixed, $C_\epsilon \asymp 1 + |\log \epsilon|$, and $\eta^2 \asymp \epsilon^3/(1 + |\log \epsilon|)$ as $\epsilon \downarrow 0$. $\square$

3.2 A bound independent of the initial mass ratio

Theorem 3.3. *There are positive constants C_{obs}, δ_m , depending only on κ, K , such that every regular cylinder of duration $C_{\text{obs}}L$ satisfies*

$$\int_0^{C_{\text{obs}}L} \left[\psi_r(s, L)^2 + \left(\frac{d}{ds} \psi(s, L) \right)^2 \right] ds > \delta_m m(0, L) \quad (40)$$

if $m(0, L) > 0$, for every constant c in $\psi = r(\phi - c)$.

Proof. Put $M_0 = m(0, L)$ and let E denote the integral on the left of (40). Suppose $E \leq \delta M_0$, with $0 < \delta \leq 1$ to be fixed below. The constant C_{obs} will also be fixed at the end; it will be at least $K + 4$. Constants denoted by C below are independent of M_0/L and δ .

If $\psi(0, L)^2 \geq M_0 L$, reverse the sign of ψ if necessary and put $q = \psi(0, L) > 0$. The positive-kernel argument in the preceding proof applies with

$$\rho = \frac{\kappa q^2}{64L^2 + \kappa q^2}.$$

Indeed, its boundary-data error is at most $\sqrt{2KLE} \leq \sqrt{2K\delta} q$, so the nonintegral term is at least $q/2$ for fixed sufficiently small δ . The resulting mass estimate is $m(s, L) \geq 4L - KE/2 > L/2$, a contradiction. Thus $\psi(0, L)^2 < M_0 L$. The energy identity and $-\log(1 - \mu) \leq \mu/(1 - \mu)$ imply

$$m(s, r) \leq \mathcal{E}_c(s, L) \leq M_* := AM_0, \quad A = 2 + \kappa^{-1}. \quad (41)$$

The other direction of the same identity gives

$$\mathcal{E}_c(s, L) \geq M_0 - E. \quad (42)$$

For the latter inequality, the absolute value of the mixed integral is at most the geometric mean of the two parts of E , and the negative square term contributes at most half of its part of E .

Write the axis in the fixed auxiliary chart as $t = \gamma(s)$. The time-to-axis assumption gives $\gamma(s) \geq 0$ for $s \geq KL$. Every incoming label on such a slice has therefore crossed $r = L$ within the cylinder. It follows that

$$K^{-1} \leq r_t \leq 1, \quad 0 < \gamma' = \frac{-(r_s)_{\text{axis}}}{(r_t)_{\text{axis}}} \leq K \quad (s \geq KL).$$

Here $-(r_s)_{\text{axis}} \leq 1$ follows by integrating r_{sr} to the outer boundary. In particular,

$$\frac{t - \gamma(s)}{K} \leq r(s, t) \leq t - \gamma(s).$$

The boundary-data part of (39) can be written as a term independent of t and an integral starting at the axis label. We denote just that first term by

$$a(s) = \psi(s, L) - \int_{\gamma(s)}^s \psi_r(\sigma, L) d\sigma, \quad s \geq KL. \quad (43)$$

Then

$$\psi(s, L) - \int_t^s \psi_r(\sigma, L) d\sigma = a(s) + \int_{\gamma(s)}^t \psi_r(\sigma, L) d\sigma.$$

The integral on the right is controlled directly by E ; the term $a(s)$ will be controlled by the radial mass equation. Differentiating (43) gives

$$a'(s) = \frac{d}{ds} \psi(s, L) - \psi_r(s, L) + \psi_r(\gamma(s), L) \gamma'(s).$$

Since γ is increasing, $0 \leq \gamma(s) \leq s \leq T$, and $\gamma' \leq K$, changing variables from s to $\gamma(s)$ yields

$$\int_{KL}^T |a'(s)|^2 ds \leq 3(1+K)E. \quad (44)$$

Also, by Cauchy–Schwarz,

$$\left| \int_{\gamma(s)}^t \psi_r(\sigma, L) d\sigma \right| \leq \sqrt{KrE}. \quad (45)$$

Set $b_0 = 32M_*$ and for now assume $L \geq 2b_0$. On $s \geq KL$, $r \geq b_0$, the mass bound implies

$$\frac{1}{2} \leq 1 - \mu \leq 1, \quad \frac{1}{2} \leq -r_s \leq 1, \quad \frac{1}{2} \leq r_t \leq 1. \quad (46)$$

For $-r_s$, integrate its radial equation from r to L . For r_t , each of the rays under consideration has entry value 1, and along that ray $-dr = (-r_s)ds$. Thus both logarithmic integrals are bounded by

$$\int_r^L \frac{2M_*}{x(x-2M_*)} dx \leq -\log(1-2M_*/r).$$

The bound for r_t is asserted only for entered rays; the restriction $s \geq KL$ ensures that all points in (46) have this property.

Apply (39) on the explicitly defined region

$$\mathcal{D} = \{(s, t) : KL \leq t \leq s \leq T, r(s, t) \geq b_0\},$$

and denote its double-integral operator by $\mathcal{K}$. Every integration point (σ, z) satisfies $t \leq z \leq \sigma \leq s$, and hence

$$r(\sigma, z) \geq r(\sigma, t) \geq r(s, t) \geq b_0.$$

Thus the operator uses only values within $\mathcal{D}$. Writing $r = r(s, t)$, (46) gives

$$r(\sigma, t) \geq r + (s - \sigma)/2, \quad V(\sigma, z) dz \leq 4M_* \frac{dr}{r^3}.$$

For every $q < 1$, first integrate in radius and then in time:

$$\begin{aligned} \mathcal{K}(r^q) &\leq \frac{4M_*}{2-q} \int_0^\infty (r + \xi/2)^{q-2} d\xi \\ &= \frac{8M_*}{(2-q)(1-q)} r^{q-1}. \end{aligned} \quad (47)$$

In the norm $\|g\|_q = \sup_{\mathcal{D}} r^{-q}|g|$, the operator norm is at most $[4(2-q)(1-q)]^{-1}$. This is at most $1/3$ for $q = 0, 1/2, -1/2$. Consequently $(I - \mathcal{K})^{-1}$ has norm at most 2 in each of these spaces, independently of L/M_* . We shall also use

$$\begin{aligned} \int_t^s \int_t^\sigma V(\sigma, z) \sqrt{s - \sigma} dz d\sigma &\leq 2M_* \int_0^\infty \frac{\sqrt{\xi}}{(r + \xi/2)^2} d\xi \\ &= \frac{2\sqrt{2}\pi M_*}{\sqrt{r}}. \end{aligned} \quad (48)$$

Let W be the solution of

$$W = 1 + \mathcal{K}W, \quad W = (I - \mathcal{K})^{-1}1.$$

This function accounts for the effect of V on the term $a(s)$ when that term is held fixed in time. Positivity and the norm bound give $1 \leq W \leq 2$. Differentiating the integral limits gives

$$W_s(s, t) = \int_t^s V(s, z)W(s, z)dz, \quad W_t(s, t) = - \int_t^s V(\sigma, t)W(\sigma, t)d\sigma.$$

Using (46) and integrating r^{-3} yields

$$1 \leq W \leq 2, \quad 0 \leq -W_r \leq \frac{CM_*}{r^2}, \quad 0 \leq W_s \leq \frac{CM_*}{r^2}. \quad (49)$$

Only integrals of V occur here; its derivatives are not needed.

Define R by the exact decomposition

$$\psi(s, t) = a(s)W(s, t) + \int_{\gamma(s)}^t \psi_r(\sigma, L)d\sigma + R(s, t). \quad (50)$$

Substitution into (39) gives

$$((I - \mathcal{K})R)(s, t) = \int_t^s \int_t^\sigma V(\sigma, z) \left[\int_{\gamma(\sigma)}^z \psi_r(\xi, L)d\xi + (a(\sigma) - a(s))W(\sigma, z) \right] dzd\sigma.$$

The first term is bounded by $CM_*r^{-1/2}\sqrt{E}$, by (45) and (47) with $q = 1/2$. For the second, (44) gives $|a(\sigma) - a(s)| \leq C\sqrt{(s - \sigma)E}$, since every $\sigma \geq t \geq KL$. Use (48), $W \leq 2$, and the inverse in the $q = -1/2$ norm to obtain

$$|R(s, t)| \leq \frac{CM_*}{\sqrt{r}}\sqrt{E}. \quad (51)$$

For the derivative estimates, differentiate the original integral equation, rather than the bound on R :

$$\psi_s(s, t) = \frac{d}{ds}\psi(s, L) - \psi_r(s, L) + \int_t^s V(s, z)\psi(s, z)dz, \quad (52)$$

$$\psi_r(s, t) = \frac{\psi_r(t, L)}{r_t(s, t)} - \frac{1}{r_t(s, t)} \int_t^s V(\sigma, t)\psi(\sigma, t)d\sigma. \quad (53)$$

In the second formula, the contribution of $a(s)W$ is $a(s)W_r$. The difference $a(\sigma) - a(s)$ is estimated using $s - \sigma \leq 2[r(\sigma, t) - r(s, t)]$ and

$$\int_r^\infty \sqrt{x - r} x^{-3} dx = \frac{\pi}{8r^{3/2}}.$$

The boundary integral in (50) contributes at most $CM_*\sqrt{E} \int_r^\infty x^{-5/2} dx$; R contributes at most $CM_*\sqrt{E} \int_r^\infty x^{-7/2} dx$. Since $M_*/r \leq 1/32$, these estimates give

$$\left| \psi_r(s, t) - \frac{\psi_r(t, L)}{r_t(s, t)} - a(s)W_r(s, t) \right| \leq \frac{CM_*}{r^{3/2}}\sqrt{E}. \quad (54)$$

The same substitution in (52), now integrating on an outgoing slice, yields

$$\left| \psi_s(s, t) - \frac{d}{ds}\psi(s, L) + \psi_r(s, L) \right| \leq C \left[\frac{M_*|a(s)|}{r^2} + \frac{M_*}{r^{3/2}}\sqrt{E} \right]. \quad (55)$$

The ray entering at time KL reaches b_0 within $2L$, since $-r_s \geq 1/2$ while its radius is at least b_0 . Thus every annulus $b_0 \leq r \leq L$ on a slice $s \geq (K+2)L$ lies in $\mathcal{D}$. On such a slice, (50), (51), and (54) give

$$\left| \frac{\theta}{\lambda} - a(s)(W_r - W/r) - \frac{\psi_r(t, L)}{r_t} + \frac{1}{r} \int_{\gamma(s)}^t \psi_r(\sigma, L) d\sigma \right| \leq \frac{CM_*}{r^{3/2}} \sqrt{E}.$$

By (49), $W/r - W_r \geq 1/r$. On $[b_0, 2b_0]$, the squared radial integral of all other terms is at most CE . For the boundary derivative use $dr = r_t dt$ and $r_t \geq 1/2$; for its primitive use (45) and $\int_{b_0}^{2b_0} r^{-1} dr = \log 2$; for the remainder use $M_*^2 \int_{b_0}^{2b_0} r^{-3} dr \leq C$. Therefore the radial mass equation and $(x+y)^2 \geq x^2/2 - y^2$ give

$$M_* \geq \frac{\kappa a(s)^2}{8b_0} - CE, \quad |a(s)| \leq CM_*. \quad (56)$$

The second assertion follows because $b_0 = 32M_*$ and $E \leq M_0 \leq M_*$.

Choose a constant $\mathcal{L} \geq 64$ and put $b = \mathcal{L}M_* < L$. Hardy's inequality gives

$$\begin{aligned} \int_b^L \frac{1}{r^2} \left| \int_{\gamma(s)}^t \psi_r(\sigma, L) d\sigma \right|^2 dr &\leq K^2 \int_{\gamma(s)}^s \frac{\left| \int_{\gamma(s)}^t \psi_r(\sigma, L) d\sigma \right|^2}{(t - \gamma(s))^2} dt \\ &\leq 4K^2 E. \end{aligned} \quad (57)$$

Here $dr \leq dt$ and $r \geq (t - \gamma(s))/K$. The primitive vanishes at its left endpoint. Its boundary term there is zero, since $\left| \int_{\gamma(s)}^{\gamma(s)+\xi} \psi_r(\sigma, L) d\sigma \right|^2 / \xi \leq \int_{\gamma(s)}^{\gamma(s)+\xi} \psi_r(\sigma, L)^2 d\sigma \rightarrow 0$; the right endpoint term in the integration-by-parts proof of Hardy's inequality is nonpositive.

Using (56), the leading term in the preceding formula for θ/λ contributes at most $CM_*^2 \int_b^L r^{-2} dr$. The boundary derivative and its primitive contribute at most CE , while the remainder contributes at most $CM_*^2 b^{-2} E$. Hence

$$m(s, L) - m(s, b) \leq C \left[\frac{M_*}{\mathcal{L}} + E \right]. \quad (58)$$

At $r = L$, definition (43) and $s - \gamma(s) \leq KL$ imply

$$\frac{\psi(s, L)^2}{2L} \leq C \left[\frac{M_*^2}{L} + E \right].$$

Since $\mathcal{E}_c(s, L) - \psi(s, L)^2/(2L) \leq m(s, L)/\kappa$, the lower energy estimate (42) gives

$$m(s, L) \geq \kappa \left[M_0 - CE - C \frac{M_*^2}{L} \right]. \quad (59)$$

Choose $\mathcal{L}$ large, then a fixed $\epsilon_0 > 0$ small, and restrict to $M_0/L \leq \epsilon_0$. By reducing δ , the last two estimates imply on every slice $s \geq (K+2)L$ that

$$m(s, b) \geq \frac{\kappa}{4} M_0. \quad (60)$$

Indeed, after division by M_0 the three error terms are bounded by constant multiples of $A/\mathcal{L}$, δ , and $A^2\epsilon_0$, respectively.

We now compare the same boundary integral at $r = b$ and $r = L$ over a late interval of outer duration at most Hb . Use the boundary measure at $r = b$, expressed in the unchanged auxiliary chart:

$$d\tau_b = -(r_s)_b ds.$$

The integral at b is

$$\begin{aligned} E_b &= \int \left[\psi_r(s, b)^2 + \left(\frac{d\psi_b}{d\tau_b} \right)^2 \right] d\tau_b \\ &= \int \left[(-(r_s)_b) \psi_r(s, b)^2 + \frac{(\psi_s - (r_s)_b \psi_r)^2}{-(r_s)_b}(s, b) \right] ds. \end{aligned}$$

Both factors of $-(r_s)_b$ follow from the derivative at fixed radius and the time measure. On $r = b$, differentiation of the equation defining that curve gives $ds/dt = (r_t)_b / (-(r_s)_b)$. Thus the boundary-data term in (54) satisfies

$$\int (-(r_s)_b) \left(\frac{\psi_r(t, L)}{(r_t)_b} \right)^2 ds = \int \frac{\psi_r(t, L)^2}{(r_t)_b} dt \leq 2E.$$

The term $d\psi(s, L)/ds - \psi_r(s, L)$ in (55) also has squared time integral at most CE , including the factor $-(r_s)_b^{-1} \leq 2$. By (56), the remaining terms in these two estimates are bounded pointwise by

$$C \left[\frac{M_*^2}{b^2} + \frac{M_*}{b^{3/2}} \sqrt{E} \right].$$

Squaring and integrating over duration at most Hb therefore gives

$$E_b \leq CE + CH \left[\frac{M_*}{\mathcal{L}^3} + \frac{E}{\mathcal{L}^2} \right]. \quad (61)$$

The constants here are independent of H and $\mathcal{L}$.

At the beginning of this interval,

$$\frac{2m(s, b)}{b} \geq \epsilon_b := \frac{\kappa}{2A\mathcal{L}}.$$

The geometric hypotheses hold on the smaller cylinder with the same κ, K . Its duration and boundary integral are measured using $d\tau_b$ as just defined. No coordinate is changed here: the identities defining E_b express that boundary integral entirely in the fixed (s, t) chart. The parametrization-independent form of Lemma 3.2, with the same constant c , therefore gives

$$E_b > e(\epsilon_b)b$$

over a boundary interval of measure $C_{\epsilon_b}b$. Since $-(r_s)_b \geq 1/2$, this interval has s -length at most Hb with $H = 2C_{\epsilon_b}$. For sufficiently large $\mathcal{L}$, (37) gives

$$H \leq C(1 + \log \mathcal{L}), \quad e(\epsilon_b)b \geq \frac{M_*}{C\mathcal{L}^2(1 + \log \mathcal{L})^2}.$$

The ratio of $CHM_*/\mathcal{L}^3$ in (61) to this lower bound is at most $C(1 + \log \mathcal{L})^3/\mathcal{L}$, which tends to zero. Choose $\mathcal{L}$ sufficiently large that this ratio is less than $1/4$. Then choose δ sufficiently small that

the two terms containing $E \leq \delta M_*/A$ sum to less than another quarter of the same lower bound. This contradicts the lower bound for E_b .

All choices are independent of M_0/L . The radius $b_0 = 32M_*$ and the constants in the integral estimates are fixed first. Choose $\mathcal{L} \geq 64$ large enough both for (60) and the last comparison; this fixes ϵ_b and H . Next reduce ϵ_0 until $L \geq 2b$, $Hb \leq L$, and the M_*^2/L term in (59) is sufficiently small. Finally fix δ to satisfy the preceding smallness requirements. The late interval starts at $(K+2)L$ and ends by $(K+3)L$. Every axis label used lies in $[0, T]$, since $\gamma(s) \geq 0$ for $s \geq KL$, and the integral equation on $\mathcal{D}$ uses only $t \geq KL$.

If $M_0/L \geq \epsilon_0$, apply Lemma 3.2 with threshold $2\epsilon_0$. Choose $C_{\text{obs}} \geq \max\{K+4, C_{2\epsilon_0}\}$. The boundary integral is then greater than $e(2\epsilon_0)L \geq 2e(2\epsilon_0)M_0$, since $M_0 \leq L/2$. Taking δ_m smaller than both this fixed coefficient and the fixed δ above proves the theorem. $\square$

3.3 Decay of the boundary energy

Corollary 3.4. *There is $\delta_E > 0$, depending only on κ, K , such that the same interval satisfies*

$$\int_0^{C_{\text{obs}}L} \left[\psi_r(s, L)^2 + \left(\frac{d}{ds} \psi(s, L) \right)^2 \right] ds \geq \delta_E \mathcal{E}_c(0, L) \quad (62)$$

for every constant c .

Proof. Write $E_0 = \mathcal{E}_c(0, L)$ and $D_0 = 1/2 + \kappa^{-1}$. The case $E_0 = 0$ is immediate. If $\psi(0, L)^2/L \leq M_0$, then $E_0 \leq D_0 M_0$, and Theorem 3.3 gives the result. Otherwise $E_0 \leq D_0 \psi(0, L)^2/L$. Suppose the boundary integral E satisfies $E \leq \delta_E E_0$. The energy identity gives $m(s, r) \leq E_0 + E/2 \leq 2E_0$ for small fixed δ_E . Choose the sign of ψ according to $\psi(0, L)$. In (39), the nonintegral term is at least $|\psi(0, L)|/2$ until the first entered ray reaches any fixed radius ρL , because its error is at most

$$\sqrt{2KLE} \leq \sqrt{2KD_0\delta_E} |\psi(0, L)|.$$

Choose $\rho \in (0, 1)$ so that $\kappa(\rho^{-1} - 1)/(16D_0) \geq 4$. The same positive-kernel and mass argument as before gives

$$m(s, L) \geq \frac{\kappa \psi(0, L)^2}{16L} (\rho^{-1} - 1) - \frac{K}{2} E \geq 4E_0 - \frac{K}{2} \delta_E E_0 > 2E_0$$

after reducing δ_E . This is a contradiction. $\square$

The choice of the scalar constant c can now be removed from the energy used in the final argument. Define

$$Q_L(s) = \frac{L}{2} \left[\left(\frac{\theta_L}{\lambda_L} \right)^2 - \log(1 - \mu_L) \right] \geq m(s, L).$$

Corollary 3.5. *There are $0 < q < 1$ and $C < \infty$, depending only on κ, K , such that every complete interval $[a, a + C_{\text{obs}}L]$ satisfies*

$$Q_L(a + C_{\text{obs}}L) \leq qQ_L(a) + CL \left[\text{TV}_{[a, a+C_{\text{obs}}L]} \left(\frac{\alpha_L}{\lambda_L} \right) \right]^2. \quad (63)$$

On every subinterval,

$$\sqrt{Q_L(s)} \leq \sqrt{Q_L(a)} + \sqrt{L/2} \text{TV}_{[a, s]} \left(\frac{\alpha_L}{\lambda_L} \right). \quad (64)$$

Proof. For the boundary estimate choose $c = \alpha(a, L)/\lambda(a, L)$. Then $\psi_r(a, L) = 0$ and $\psi(a, L)/L = -\theta(a, L)/\lambda(a, L)$, so $\mathcal{E}_c(a, L) = Q_L(a)$. Write V_a for the total variation in (63). Since $|\psi_r(s, L)| \leq V_a$, the preceding corollary gives

$$\int_a^{a+C_{\text{obs}}L} \left(\frac{d}{ds} \psi(s, L) \right)^2 ds + C_{\text{obs}} L V_a^2 \geq \delta_E Q_L(a).$$

Along $r = L$,

$$\frac{d}{ds} \left(\frac{\theta_L}{\lambda_L} \right) = \frac{d}{ds} \left(\frac{\alpha_L}{\lambda_L} \right) - \frac{d}{ds} \phi(s, L).$$

Combining this with the boundary mass identity and $L d\phi(s, L)/ds = (r\phi_s)_L + \theta_L/\lambda_L$ yields

$$\frac{dQ_L}{ds} + \frac{L^2}{2} \left(\frac{d}{ds} \phi(s, L) \right)^2 = L \frac{\theta_L}{\lambda_L} \frac{d}{ds} \left(\frac{\alpha_L}{\lambda_L} \right). \quad (65)$$

Since $|\theta_L/\lambda_L| \leq \sqrt{2Q_L/L}$, for $\varepsilon > 0$ this implies

$$\frac{d}{ds} \sqrt{Q_L + \varepsilon} \leq \sqrt{L/2} \left| \frac{d}{ds} \left(\frac{\alpha_L}{\lambda_L} \right) \right|.$$

Integrate and let $\varepsilon \downarrow 0$ to prove (64), including at zeros of Q_L . The absolute value of the integrated right side of (65) is therefore at most

$$\sqrt{2LQ_L(a)} V_a + L V_a^2.$$

Moreover, $d\psi(s, L)/ds = L d\phi(s, L)/ds$. Combining the lower bound for its squared integral with (65), and using

$$\sqrt{2LQ_L(a)} V_a \leq \frac{\delta_E}{4} Q_L(a) + \frac{2}{\delta_E} L V_a^2,$$

gives (63) with

$$q = 1 - \delta_E/4, \quad C = 1 + C_{\text{obs}}/2 + 2/\delta_E.$$

We may take $0 < \delta_E \leq 1$. The definition of Q_L and identity (65) are independent of the constant c used on this particular boundary interval. $\square$

4 Continuation at the central endpoint

By the preceding estimates, there are constants independent of the radius such that

$$\ell \leq \lambda \leq \Lambda, \quad 1 - \mu \geq \kappa > 0, \quad \left| \frac{\theta}{\lambda} \right| + |r\alpha'| \leq C, \quad \left| \frac{\theta}{\lambda} \right|^3 \leq C\mu. \quad (66)$$

Since $-\nu = \lambda$ on the axis and $\partial_r(-\nu) \geq 0$, we also have $-\nu \geq \ell$ in the fixed coordinates. Write

$$R(u) = r(u, 0), \quad M(u) = m(u, 0).$$

Then

$$\ell(-u) \leq R(u) = \int_u^0 \lambda(u, v) dv \leq \Lambda(-u). \quad (67)$$

On a regular cylinder $0 \leq r \leq b$, set, as a function of the original outgoing coordinate u ,

$$Q_b(u) = \frac{b}{2} \left[\left(\frac{\theta}{\lambda} \right)_b^2 - \log(1 - \mu_b) \right] \geq m_b. \quad (68)$$

Use the boundary measure $d\tau_b = (-\nu_b)du$ from (28). Corollary 3.5 gives constants $H > 0$, $0 < q < 1$, and $C > 0$ such that, for any regular cylinder interval $[a_0, a_1]$ with

$$\int_{a_0}^{a_1} (-\nu_b) du = Hb,$$

we have

$$Q_b(a_1) \leq qQ_b(a_0) + Cb \left[\text{TV}_{[a_0, a_1]} \left(\frac{\alpha}{\lambda} \right)_b \right]^2. \quad (69)$$

On every regular subinterval, with no prescribed duration,

$$\sqrt{Q_b(a_1)} \leq \sqrt{Q_b(a_0)} + \sqrt{b/2} \text{TV}_{[a_0, a_1]} \left(\frac{\alpha}{\lambda} \right)_b. \quad (70)$$

These are boundary identities and estimates in the fixed chart. Total variation is invariant under increasing parametrizations of the boundary, which justifies the use of the original u -endpoints. Moreover, (66) implies $Q_b \leq Cb$. All constants in this section are independent of the parameter ε introduced below.

4.1 Decay of the mass ratio on the limiting cone

The integral in (4) is finite also on $v = 0$. Indeed, continuity on compact regular segments as $v \uparrow 0$ gives the same bound on every segment $u_i \leq u \leq u_0 < 0$, and monotone convergence then gives the bound on the whole cone. Since R decreases to zero, a change of variable from u to $r = R(u)$ yields

$$\int_0^{R(u_i)} \frac{\mu}{(1-\mu)r} dr < \infty.$$

Along this cone, m is nondecreasing as a function of the outward radius, because $m_u \leq 0$ and $r_u < 0$. If $\mu(r_j) \geq \eta > 0$ for a sequence $r_j \downarrow 0$, then

$$\mu(r) \geq \frac{r_j}{r} \mu(r_j) \geq \frac{\eta}{2} \quad (r_j \leq r \leq 2r_j).$$

A subsequence can be chosen so that these intervals are disjoint. Each contributes at least $(\eta/2) \log 2$ to the preceding integral, which is impossible. This is the monotonicity argument of [2, p. 190, Lemma 2], in the present coordinates. Thus

$$M(u) = o(R(u)) = o(-u), \quad \partial_u M \leq 0. \quad (71)$$

4.2 The trace on a cylinder with small exterior mass ratio

Suppose that $r = b$ is contained in the regular region on an interval starting at u_0 , and that $\mu \leq \varepsilon$ at every point of this interval with $r \geq b$. An incoming characteristic ending at (u, b) remains in $r \geq b$ when followed backwards to u_0 . The differentiated field equation is

$$\begin{aligned} \partial_u(r\alpha') &= \frac{-\nu}{r} \left(\frac{2\mu}{1-\mu} - 1 \right) (r\alpha') \\ &\quad + \frac{-\nu}{(1-\mu)r} \frac{\theta}{\lambda} \left[\left(\frac{\theta}{\lambda} \right)^2 - 3\mu \right]. \end{aligned} \quad (72)$$

By (66),

$$\frac{1}{1-\mu} \left| \frac{\theta}{\lambda} \left[\left(\frac{\theta}{\lambda} \right)^2 - 3\mu \right] \right| \leq C\mu.$$

The homogeneous propagator from t to u is

$$\frac{r(u, v)}{r(t, v)} \exp \left(2 \int_t^u \frac{\mu(-\nu)}{(1-\mu)r}(s, v) \, ds \right).$$

Writing $r_i = r(u_0, v)$ and using the uniform integral bound, we obtain

$$\begin{aligned} |r\alpha'(u, v)| &\leq C \frac{r(u, v)}{r_i} + C\varepsilon r(u, v) \int_{u_0}^u \frac{-\nu(s, v)}{r(s, v)^2} \, ds \\ &\leq C \left(\frac{r(u, v)}{r_i} + \varepsilon \right), \end{aligned}$$

where the last step uses $r_u = \nu$. At the endpoint $r = b$, $r_i \geq b + \ell(u - u_0)$. Differentiation along the cylinder gives

$$\frac{1}{-\nu_b} \frac{d}{du} \left(\frac{\alpha}{\lambda} \right)_b = \frac{1}{b} \left[b\alpha'_b + \frac{\mu_b}{1-\mu_b} \left(\frac{\theta}{\lambda} \right)_b \right]. \quad (73)$$

Consequently, after $u - u_0 \geq b/(\ell\varepsilon)$,

$$\text{TV} \left(\frac{\alpha}{\lambda} \right)_b \leq C\varepsilon \quad (74)$$

on every subsequent interval $[a_0, a_1]$ contained in the cylinder with $\int_{a_0}^{a_1} (-\nu_b) du \leq Hb$.

4.3 Decay of the boundary energy

Starting after this waiting period, apply (69) on consecutive intervals $[a_k, a_{k+1}]$ chosen by

$$\int_{a_k}^{a_{k+1}} (-\nu_b) \, du = Hb.$$

At their endpoints,

$$Q_b(a_k) \leq Cq^k b + C\varepsilon^2 b.$$

Choose $\varepsilon > 0$ sufficiently small and then $N = N(\varepsilon)$ sufficiently large that the two terms on the right are each at most $\varepsilon b/64$ for $k \geq N$. For any point between consecutive endpoints, (70) and (74) give, after decreasing the permitted upper bound for ε if necessary,

$$Q_b \leq 2(\varepsilon b/32) + C\varepsilon^2 b \leq \varepsilon b/8. \quad (75)$$

The same estimate holds on a final interval whose boundary measure is less than Hb . Thus (75) holds for the remainder of any such cylinder after the waiting period and N complete intervals.

4.4 Selection of the cylinder

Suppose that (5) fails. Choose ε small enough for the preceding estimates and strictly below the resulting positive full-triangle $\limsup \mu$. There are then points with $\mu > \varepsilon$ at arbitrarily late outgoing times. By (71), we may restrict to a fixed late subtriangle on which $\mu(u, 0) \leq \varepsilon$. Define

$$b_*(u) = \sup \{ r(t, v) : (t, v) \in \mathcal{Q}, \, t \geq u, \, \mu(t, v) > \varepsilon \}.$$

The set is nonempty for every $u < 0$. Radial mass monotonicity and $M(t) \leq M(u)$ for $t \geq u$ imply

$$0 < b_*(u) \leq \frac{2M(u)}{\varepsilon} = o(-u). \quad (76)$$

In particular, b_* is nonincreasing and tends to zero.

We record two consequences of regularity needed below. Every point with $\mu > \varepsilon$ lies away from both the axis and $v = 0$. It therefore has a neighborhood in (u, r) coordinates contained in the regular region. To prove right-continuity of b_* at u , choose, for any $\delta > 0$, such a point at a time $t \geq u$ with radius greater than $b_*(u) - \delta$. If $t > u$, this point remains in the defining set for all sufficiently small increases of u . If $t = u$, continuity preserves $\mu > \varepsilon$ at the same radius for sufficiently small increases of t . In either case,

$$\lim_{s \downarrow u} b_*(s) \geq b_*(u) - \delta.$$

Monotonicity and the arbitrariness of δ prove right-continuity. Moreover, for every $t \geq u$, all available radii $r \geq b_*(u)$ satisfy $\mu(t, r) \leq \varepsilon$. For $r > b_*(u)$ this follows from the definition; at equality, a strict inequality $\mu > \varepsilon$ would persist at a slightly larger radius and give a contradiction.

Choose $b_0 > 0$ smaller than the value of b_* at the initial time of this subtriangle. Set $b_n = 2^{-n}b_0$ and let u_n be the first time at which $b_*(u) \leq b_n$. These times are finite, since $b_*(u) \rightarrow 0$, and the threshold is attained by right-continuity. They are nondecreasing, with repeated values allowed, and $u_n \uparrow 0$: for each fixed $u < 0$, positivity of $b_*(u)$ implies $b_n < b_*(u)$ and hence $u_n > u$ for large n . Thus

$$b_*(u) \leq b_n \quad (u \geq u_n), \quad b_*(u) > b_n/2 \quad (u_n \leq u < u_{n+1}).$$

For $u < u_n$ we have $b_*(u) > b_n$. Letting $u \uparrow u_n$ in (76), using continuity of M at the regular point $(u_n, 0)$, gives

$$b_n \leq \frac{2M(u_n)}{\varepsilon} = o(-u_n).$$

It follows that

$$\limsup_{n \rightarrow \infty} \frac{u_{n+1} - u_n}{b_n} = \infty. \quad (77)$$

Indeed, an eventual bound $u_{n+1} - u_n \leq Cb_n$ would imply

$$-u_n = \sum_{j \geq n} (u_{j+1} - u_j) \leq C \sum_{j \geq n} b_j = 2Cb_n,$$

contrary to $b_n = o(-u_n)$.

Choose indices along which the quotient in (77) tends to infinity. For all sufficiently large such indices, the cylinder $0 \leq r \leq b_n$ exists throughout $[u_n, u_{n+1}]$, because R is decreasing and

$$R(u_{n+1}) \geq \ell(-u_{n+1}) \gg b_{n+1} = b_n/2.$$

Its exterior satisfies $\mu \leq \varepsilon$ throughout this interval. The waiting period has original coordinate length $b_n/(\ell\varepsilon)$, and, since $-\nu_b \geq \ell$, the next N intervals of boundary measure Hb_n require at most NHb_n/ℓ further units of u . By (77), both fit before u_{n+1} , with a nonempty interval remaining.

Choose $u_* < u_{n+1}$ after these intervals. Since $b_*(u_*) > b_n/2$, its defining supremum supplies a point (t, r_*) such that

$$t \geq u_*, \quad b_n/2 < r_* \leq b_n, \quad \mu(t, r_*) > \varepsilon.$$

In fact $t < u_{n+1}$, because $b_*(u_{n+1}) \leq b_n/2$. Radial mass monotonicity then gives

$$m(t, b_n) \geq m(t, r_*) > \frac{\varepsilon b_n}{4}.$$

On the other hand, (75) applies at t , and hence

$$\frac{\varepsilon b_n}{4} < m(t, b_n) \leq Q_{b_n}(t) \leq \frac{\varepsilon b_n}{8},$$

a contradiction. This proves (5).

4.5 Application of Christodoulou's extension theorem

Equation (5) implies that the supremum of μ over a past subtriangle tends to zero as its initial outgoing slice approaches the endpoint. This is the small-mass hypothesis of [1, p. 1172, Theorem 5.1 and (5.1)].

The initial gauge has already been fixed. To use the labels in [1, p. 1172, Theorem 5.1], make only the translation

$$U = u - u_i, \quad V = v - u_i, \quad v_* = -u_i.$$

The axis remains diagonal, the initial cone becomes $U = 0$ with $r(0, V) = V/2$, and the endpoint has labels (v_*, v_*) . The translation introduces no factors in $\lambda, \nu, \theta, \zeta$, or α . In particular the translated initial datum is C^1 on a neighborhood of $[0, v_*]$. Extend it to a C^1 function on $[0, \infty)$ by a cutoff supported in a slightly larger initial-data interval and equal to one near $[0, v_*]$.

For every $0 < v_0 < v_*$, the given solution on the corresponding smaller closed triangle is in the C^1 class of [1, p. 1157]: regularity and compactness give the required lower bounds for λ and ν there. These restrictions provide the smaller-triangle solutions with the extended datum required by Theorem 5.1. The full-triangle limit already proved gives

$$\lim_{a \uparrow v_*} \sup_{a < U < V < v_*} \mu(U, V) = 0.$$

Thus Theorem 5.1 supplies a C^1 solution on a larger triangle. Characteristic uniqueness [1, p. 1148, Theorem 3.1] identifies it with the original solution on every smaller triangle, and continuity gives agreement on the regular part of the limiting cone. Translating back gives the desired C^1 continuation across $(0, 0)$.

5 What this criterion supplies for weak cosmic censorship

The contrapositive of Theorem 1.1 is a statement about a family of cones: a nonextendible central endpoint satisfying the other hypotheses cannot have uniformly bounded complete incoming blueshift integrals. It does not assert that the integral on its limiting cone $v = 0$ is infinite. The estimates in Sections 2 and 4 use the uniform bound on the other cones, in particular in the characteristic propagator for $r\alpha'$.

To compare the statements precisely, Christodoulou defines the blueshift exponent γ in [2, p. 189, (2.1)]. On the distinguished cone $s = 0$, its radius is $r = ae^{-t}$ and $d\gamma = \mu(1 - \mu)^{-1}dt$; see [2, p. 187, (1.13), and p. 189, (1.26)]. Since $dt = -dr/r$, boundedness of γ is equivalent to

$$\int_{u_i}^0 \frac{\mu(-\nu)}{(1 - \mu)r}(u, 0) du < \infty.$$

Lemma 2, p. 190, proves decay of μ on this cone. The paragraph following it, continued on p. 191, invokes the extension criterion and proceeds with unbounded γ . The extension theorem actually requires smallness on a whole triangle, as stated in [1, p. 1172, Theorem 5.1]. The present proof obtains that interior control under the stronger uniform hypothesis. It does not prove the reverse implication from finite terminal-cone blueshift to that uniform hypothesis.

This distinction matters for the hypotheses of the instability results. Both [2, p. 192, Theorem 2.1, and pp. 213–214, Theorem 3.1] explicitly assume unbounded γ . The exceptional set defined on p. 214 includes that condition, and the positive-codimension conclusion of Theorem 4.1, p. 216, is stated in $BV \cap L^1(\mathbb{R})$, with the data variable and logarithmic radial coordinate specified on p. 214, (4.1)–(4.2). It is a statement about disjoint two-parameter affine families through exceptional data. It should be distinguished from an open dense genericity statement in a differentiable topology.

There is a separate regularity issue in applying this criterion to such perturbations. In the instability paper the symbol α means $\partial_r(r\phi)$; it is α/λ in the notation used here, and differs by a factor of two from our initial datum when $r = v/2$. Also, the first perturbation direction in [2, p. 215, (4.5)] vanishes to the left of zero and has right limit one there; the second direction is chosen absolutely continuous. Thus the full two-parameter family as written is a BV family, not a family of C^1 data. Neither those perturbations nor arbitrary AC data are covered just by the C^1 continuation argument above.

For comparison, [3, Section 1.4.1, Theorem 1.4 and the following discussion] gives an AC formulation of the four-dimensional instability result and explicitly identifies the missing published passage to unbounded terminal-cone blueshift. Its own main theorem concerns circular symmetry in $2 + 1$ dimensions with negative cosmological constant; the mass-gap argument supplying its blueshift input is in Section 1.3.3. That argument does not supply the missing four-dimensional implication here.

To use the present result in the original four-dimensional WCC argument, one would still need either to derive uniform interior-cone control from bounded terminal-cone blueshift, to derive the required triangle smallness directly from that single-cone hypothesis, or to adapt the instability argument to the unbounded family supplied by our contrapositive. A formulation for AC or BV data would additionally require estimates and a continuation theorem in that class. The perturbation, genericity, and global causal conclusions remain separate parts of the WCC argument. The criterion proved here resolves none of these additional steps by itself.

A Formal verification and provenance

Credit and scope. This work was prompted by **Serban Eugen Cicortas**. The manuscript and Lean formalization were developed with OpenAI Codex under his prompts and direction. This sharing edition adds attribution and verification information to the manuscript used for formalization; its mathematical body is unchanged.

The formalization establishes the uniformly bounded family-of-cones blueshift criterion: uniform mass-ratio decay at the centre and the stated C^1 continuation. Lean checks the encoded theorem from its encoded hypotheses. This does not constitute automatic verification of every sentence of the manuscript, nor does it extend the result to a single terminal-cone hypothesis, arbitrary AC/BV data, or the remaining WCC conclusions.

Explicit imported assumptions. As intended, Christodoulou’s BV results are inputs to the formal theorem, not results proved within this project. These comprise the transport identity (4.10a), the weighted derivative identity (5.2), the extension input from Theorem 5.1, and the uniqueness input from Theorem 3.1 of [1]. The primitive field equations and the geometric, regularity, gauge, datum, and blueshift conditions also appear among the formal hypotheses. The exact definitions and quantifiers can be inspected in [StatementInterface.lean](#).

Proof and verification record. The proof is archived in [the extension-principle-lean repository](#), at the fixed commit

e78705df465b8bc1c4d4a2fdd2f13cc17a6b42bd.

The final theorem is `uniformly_bounded_blueshift_and_continuation_proved` in namespace `ExtensionPrinciple.Proofs`; its proof is in [Proofs/Main.lean](#). The separate decay theorem is in [Proofs/Decay.lean](#).

The committed [verification record](#) reports successful compilation of 513 accepted proof modules and exact-type checks for the eight frozen keystones and both root propositions. The final axiom checks list only `propext`, `Classical.choice`, and `Quot.sound`: no `sorryAx` or additional project axioms. The BV inputs remain explicit theorem hypotheses. These checks were performed on the server before the completed proof was committed; this edition cites that archived verification record.

Reproducibility and access. The repository pins Lean 4.35.0-rc3 and Mathlib revision `c55e6e786f49471c72fbddbec5415808896aec1e`, together with its transitive dependencies. Its `README.md` documents the build, and `scripts/verify_campaign.py` performs the campaign checks. All links above refer to the fixed commit rather than a moving branch. The repository is currently private; readers need access from its owner to inspect or rebuild the proof.

The original mathematical source is `bounded-blueshift-extension.tex` in that commit. Its SHA-256 fingerprint is

d72ca1ad418e9cde39301519fcbbff0bc7
cd14fad216275f295734231726ef37.

The fingerprint identifies the original source, not this sharing edition with its added front matter and appendix.

References

- [1] D. Christodoulou, *Bounded Variation Solutions of the Spherically Symmetric Einstein–Scalar Field Equations*, Communications on Pure and Applied Mathematics **46** (1993), 1131–1220.
- [2] D. Christodoulou, *The instability of naked singularities in the gravitational collapse of a scalar field*, Annals of Mathematics, Second Series, **149** (1999), no. 1, 183–217.
- [3] S. Cicortas, *Weak cosmic censorship for the circularly symmetric Einstein–scalar field system in $2 + 1$ dimensions*, [arXiv:2605.19143v2](https://arxiv.org/abs/2605.19143v2) (9 June 2026).
- [4] D. Christodoulou and A. S. Tahvildar-Zadeh, *On the asymptotic behavior of spherically symmetric wave maps*, Duke Mathematical Journal **71** (1993), no. 1, 31–69. [doi:10.1215/S0012-7094-93-07103-7](https://doi.org/10.1215/S0012-7094-93-07103-7).
- [5] J. Singh, *Regimes of (in-)stability for self-similar naked singularities*, Ph.D. dissertation, Princeton University, May 2025. [Princeton repository](https://princeton-repository.org/).
- [6] J. Liu and J. Li, *A robust proof of the instability of naked singularities of a scalar field in spherical symmetry*, Communications in Mathematical Physics **363** (2018), 561–578. [doi:10.1007/s00220-018-3157-1](https://doi.org/10.1007/s00220-018-3157-1).